

UNIT 1: Introduction to MERN Stack

1. Apply ES6 arrow functions to convert a traditional function.
2. Illustrate the request-response flow in a MERN application.
3. Analyse the role of MongoDB in handling dynamic and unstructured data.
4. Differentiate between Node.js and traditional server technologies.
5. Evaluate the impact of ES6 features on modern web development.
6. Design a simple architecture of a MERN stack application.

UNIT 2: Backend with Node.js & Express

1. Analyse the flow of request handling in Express (route → middleware → controller → response).
2. Differentiate between synchronous and asynchronous operations in Node.js.
3. In a User Management System, explain the role of controllers and how they separate business logic from routes.
4. Identify the role of controllers in API design.
5. Use middleware to log incoming requests.
6. Examine the purpose of middleware in backend development.
7. Illustrate error handling in Express applications.

UNIT 3: Frontend with React

1. Use the useState hook to manage component data.
2. Demonstrate passing data using props between components.
3. In a Student/Employee Registration Form, implement form handling in React and display submitted data.
4. Analyse the difference between state and props.
5. Examine the working of the useEffect hook.
6. In a Counter/Temperature Control Application, use the useState hook to perform the following operations:
 - Increment the value when the "+" button is clicked
 - Decrement the value when the "-" button is clicked
 - Reset the value to the default (initial) state when the reset button is clicked
7. Implement navigation using React Router.
8. Illustrate form handling using controlled components.
9. Compare functional and class components.

UNIT 4: MongoDB Database

1. In a Student Management System, the student collection stores student details such as name, age, course, and marks. Write MongoDB queries to perform the following operations:
 - Insert five student records into the collection.
 - Display all student records.
 - Update the marks of a particular student.
 - Delete a specific student record based on name or ID.
2. Analyse differences between SQL and MongoDB.
3. Compare embedded vs referenced data models.
4. Evaluate schema design strategies for performance optimisation.
5. In an Online Food Delivery System, the order collection stores details such as customerName, product, quantity, and totalAmount.

Write a MongoDB aggregation query to perform the following operations:

- Group orders by customerName
 - Calculate the total purchase amount for each customer
 - Count the total number of orders placed by each customer
6. Design a MongoDB schema for an Online Food Delivery Application (customers, restaurants, orders, payments, delivery).

UNIT 5: Full-Stack Integration & Deployment

1. Integrate frontend with backend using API calls.
2. Use environment variables in a Node.js application.
3. Demonstrate API testing using Postman.
4. Illustrate deployment steps for a MERN application.
5. Compare development vs production environments.
6. Examine the role of CI/CD in deployment.
7. Design a deployment architecture for a MERN application.